

LINFO1115 Final exam, June 1, 2021

TOTAL POINTS

17.5 / 30

QUESTION 1

1 Definition of Braess's Paradox 1.5 / 2

- 0 pts Correct
- 2 pts Incorrect
- ✓ - 0.5 pts Mainly correct but something missing (adding road, performance at equilibrium, not always worse)
- 1 pts Partially correct but substantial missing concepts

QUESTION 2

2 Example of Braess's Paradox 3 / 3

- ✓ - 0 pts Correct
- 3 pts Incorrect
- 2 pts Partial solution with incorrect reasoning and no time numbers
- 1 pts Partial solution not comparing actual times
- 1 pts Form is correct, but computation is wrong
- 2 pts Partial solution with no clear comparison

QUESTION 3

Penalty Kick game 5 pts

3.1 Principle of indifference 1 / 2

- 0 pts Correct
- 2 pts Incorrect
- ✓ - 1 pts Incorrect explanation for principle of indifference
- 1 pts Incorrect explanation for mixed-strategy equilibrium
- 0.5 pts Partially correct for principle of indifference
- 0.5 pts Partially correct for mixed-strategy equilibrium

3.2 Mixed-strategy equilibrium for Penalty-Kick game 3 / 3

✓ - 0 pts Correct

- 3 pts Incorrect
- 0.5 pts Calculation error in Kicker and/or Keeper probability computation
- 1 pts Only mentions one probability
- 2 pts Gives payoffs but does not equate them to compute probabilities
- 2 pts Wrong payoff computations are equated
- 0.5 pts No simplification of payoff equations

QUESTION 4

4 Adjacency matrix and random walk matrix

3 / 3

✓ - 0 pts Correct

- 3 pts Incorrect
- 0.5 pts Incomplete explanation for A
- 1 pts Wrong for A^2 and A^3
- 1 pts Wrong for M
- 0.5 pts Incomplete explanation for A^2 and A^3
- 0.5 pts Incomplete explanation for M
- 1 pts Wrong for A

QUESTION 5

5 Page Rank in matrix form 0 / 4

- 0 pts Correct
- 1 pts Teleportation matrix not defined.
- ✓ - 4 pts Incorrect
- 1 pts Eigenvector equation not defined.
- 1 pts Modified eigenvector equation not defined.
- 1 pts Dead ends and spider traps not defined.
- 0.5 pts Eigenvector equation partially correct
- 0.5 pts Teleportation matrix partially correct.
- 0.5 pts Modified eigenvector equation partially correct

QUESTION 6

Link farms 7 pts

6.1 Link farm definition 2 / 3

- 0 pts Correct
- 3 pts Incorrect
- 1 pts Missing ingoing links (accessible / inaccessible).
- ✓ - 1 pts Missing pages owned by link farm, with bidirectional links.
- 1 pts Missing correct textual explanation (including purpose).

6.2 Operation of a link farm 0 / 4

- 0 pts Correct
- ✓ - 4 pts Incorrect
- 1 pts Only explains partially the two effects, namely links from accessible pages and links from the farm.
- 1 pts Does not discuss setting up a company.
- 1 pts Does not explain how to increase the effect (either with equation or by discussion), in particular effect of teleportation on link farm pages.

QUESTION 7

Direct-benefit cascades 6 pts

7.1 Direct benefit effects 2 / 2

- ✓ - 0 pts Correct
- 1 pts Wrong definition of direct benefit.
- 1 pts Wrong example of direct benefit.

7.2 Tipping points 2 / 4

- 0 pts Correct
- 4 pts Incorrect
- ✓ - 1 pts Incorrect reservation price definition.
- ✓ - 1 pts Incorrect benefit function definition.
- 1 pts Incorrect explanation of tipping point.
- 3.5 pts Only partial explanation of tipping point
- 1 pts No graph with equilibria.

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(Note: Q1, Q2, Q3 taken together correspond to the midterm.)

Q1 Definition of Braess's Paradox. [2pts]

Give a precise definition of Braess's Paradox for traffic networks.

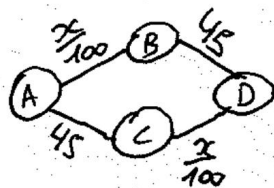
Un Braess paradox dans un traffic network correspond au fait que si l'on ajoute une route (une stratégie) alors la situation est empirée.

Donc l'ajout d'une ressource m'améliore pas toujours la situation

Q2 Example of Braess's Paradox. [3pts]

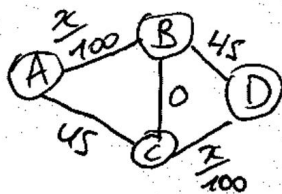
Give a concrete example of a traffic network that shows Braess's Paradox and explain how it occurs.

Situation de départ:



Nous avons ici un équilibre de Nash. Les deux routes vont être utilisées. Si le nombre de joueurs est de 4000, alors le coût causé par l'équilibre est de $\frac{2000}{100} + 45 = 65$ pour les 2 routes.

Rajoutons une route B-C:



Ici nous constatons que l'équilibre de Nash n'est plus celui d'avant.

Maintenant tous les joueurs vont emprunter la route A-B-C-D qui a un coût de $\frac{4000}{100} + 0 + \frac{4000}{100} = 80$. Sans une régulation de l'état aucune autre route est meilleure.

Q3 Penalty Kick game.

Given the Penalty Kick game we saw in the course with the following payoff matrix:

	Keeper	
	<i>Dive left</i>	<i>Dive right</i>
Kicker	<i>Kick left</i>	0.58, -0.58 0.95, -0.95
	<i>Kick right</i>	0.93, -0.93 0.70, -0.70

This game models penalty kicks in soccer. The payoffs are taken from an analysis of real soccer games. This game has no pure Nash equilibrium, but there is a mixed-strategy Nash equilibrium.

Q3.1 Principle of indifference. [2pts]

Explain the principle of indifference which is used to compute mixed-strategy Nash equilibria. In your explanation, define what is a mixed-strategy equilibrium.

Ici il n'y a aucun pure Nash et donc les joueurs sont indifférent et vont choisir des probabilités plutôt que des ~~pure~~ stratégies.

On va donc définir les mixed-strategy Nash equilibria qui vont définir les probabilités de choisir une stratégie ou une autre en restant indifférent.

Q3.2 Mixed-strategy equilibrium for Penalty-Kick game. [3pts]

Use principle of indifference to compute the mixed-strategy Nash equilibrium for the Penalty-Kick game.

$$p = \text{Dive left} \quad 0.58p + 0.95(1-p) = 0.93p + 0.70(1-p)$$

$$0.60p = 0.25$$

$$p = \frac{10}{24}$$

$$q = \text{Kick left} \quad -0.58q - 0.93(1-q) = -0.95q - 0.70(1-q)$$

$$\frac{1}{2}q = 0.23$$

$$q = 0.46$$

Donc le mixed strategy vaut $\left(\overset{p}{\frac{10}{24}}, \overset{q}{0.46}\right)$

(Les calculs ont été fait sans calculatrice)

Q4 Adjacency matrix and random walk matrix. [3pts]

Define the adjacency matrix A for a graph $G=(V,E)$. Explain what is the meaning of A^2 and A^3 in terms of graph theory. Define also the random walk matrix M (also known as the transition matrix) in terms of A .

- La matrice d'adjacence définit un graph $G=(V,E)$
 $A[i][j]$ vaut un si il y a un edge partant du noeud i qui rejoint le noeud j . 0 sinon.
- A^2 et A^3 permet d'avoir une matrice décrivant combien de chemin de longueur n d'un noeud i à un noeud j . n étant la puissance et $A^n[i][j]$ le nb de chemin de longueur n de i à j .
- La transition matrix est défini comme $M = DA$
où A est la matrice d'adjacence et D la matrice diagonale où chaque nombre vaut $\frac{1}{m}$, m étant nb de edge par noeud

Q5 Page Rank algorithm in matrix form. [4pts]

Google Page Rank can be seen as the principal eigenvector of the transition matrix of the Web graph. Explain this statement and give the equation corresponding to the statement. However, this is not enough because of the problems of dead ends and spider traps. Explain these problems. To fix these problems we add the concept of teleportation. Explain this and show how the Page Rank equation is modified to add teleportation.

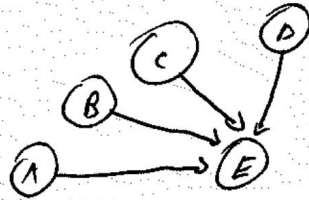
- Nous avons donc la matrice $M = DA$
- Pour la teleportation, dans une probabilité β nous restons sur le noeud et dans une probabilité $(1-\beta)$ nous nous téléportons afin d'éviter spider trap et dead ends.

Q6 Link farms

Q6.1 Link farm definition. [3pts]

Explain what is a link farm and give the graph structure of a typical link farm.

Le link farm est l'opération de faire le max de lien entrant afin d'avoir un pagerank plus élevé. Par exemple une page peut en payer d'autres pour avoir plus de lien entrant



- Ici fait du link farm

Q6.2 Operation of a link farm. [4pts]

Using the random walk definition of Page Rank, explain the operation of a link farm. How does the link farm achieve its effect? Could you set up a company to provide link farm services to third parties?

Q7 Direct-benefit cascades.

Q7.1 Direct-benefit effects. [2pts]

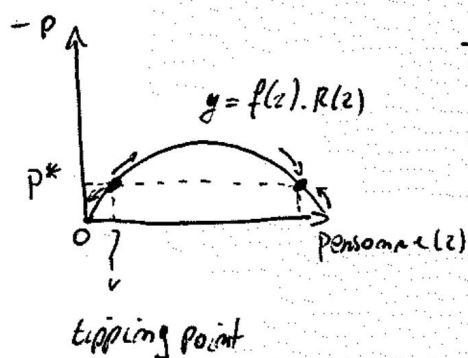
Explain what are direct-benefit effects in an economic model with a product that is bought by consumers. Give at least one example of a product with a direct-benefit effect.

- Le direct benefit effect décrit que plus un produit sera utilisé par la population et plus ce produit a de la valeur. Cette valeur ^{ajoute} dépendant du nombre de personnes n'est pas intrinsèque à l'objet.
- Le téléphone, plus il y a de personnes et plus il a de la valeur car je peux appeler plus de personnes.

Q7.2 Tipping points. [4pts]

Define enough of the formal model for direct-benefit effects to explain how unstable equilibria occur (you need to define two functions). In your model, show the equilibria that occur and explain for each one whether it is stable or unstable. Explain how the unstable equilibrium gives the effect of a "tipping point": explain what is a tipping point and how the unstable equilibrium achieves a tipping point.

- formal model : $y = f(z) \cdot R(z)$
 - $f(z)$ est $1-z$, moins c'est cher et plus je veux acheter
 - $R(z)$ vaut z , plus il y a de personnes et plus je veux acheter



- Nous avons 3 équilibres
 - 0 celui-ci est stable
Personne n'achète et ne paye
 - tipping point : ce point est instable, peu de gens et ils paient trop cher.

- Le tipping point est donc instable et il faut passer ce point pour que l'effet social nous ramène vers P^* .
Si on ne parvient pas à le passer alors l'effet social nous ramène vers 0 (Downward vs ^{upward} _{price})
- Ensuite nous avons le deuxième $y = P^* = f(z^*) \cdot R(z^*)$ celui-ci est stable et désire.